

Module 1: Java Fundamentals & Basics

- Explain OOP principles (Encapsulation, Inheritance, Polymorphism).
- Explain lexical issues and data types in Java.
- Write a Java program to add two matrices or sort an array.
- Explain type conversion, type casting, and operator types (shift operators <<, >>, >>>).

Module 2: Classes, Objects, and Methods

- Define constructors and explain types (default and parameterized).
- Distinguish between method overloading and method overriding.
- Explain the use of this keyword, static variables, and static methods.
- Write a program to implement Stack operations or a recursive Fibonacci/Factorial function.
- Explain call by value and call by reference.

Module 3: Inheritance and Interfaces

- Explain inheritance and its types in Java.
- Discuss dynamic method dispatch with an example.
- Differentiate between abstract classes and interfaces.
- Write a program implementing an interface.

Module 4: Packages and Exception Handling

- Create and use user-defined packages.
- Explain exception handling keywords (try, catch, throw, throws, finally).
- State the difference between throw and throws.
- Write a program for custom exception handling. [\[1\]](#)

Module 5: Multithreading

- Explain thread creation using the Thread class and Runnable interface.
- Discuss the thread lifecycle and priorities.
- Explain thread synchronization with a suitable example.
- Explain wrapper classes and autoboxing/unboxing.